

**Mathematical Optimization and Operational Strategies in Tiffin Delivery
Services.
(Semester III)**

A Report Submitted to
SVKM's
Narsee Monjee Institute of Management Studies (NMIMS)

**as a part of Semester III
of
Bachelor of Science (Applied Mathematical Computing
By**

A001 Nafsiha Ahmed

A014 Hibah Patanwala

A020 Krishna Vyas

Under the Guidance of

Dr. Shikha Gaur

SVKM's

NMIMS

Deemed to be UNIVERSITY

Nilkamal School of Mathematics, Applied Statistics and Analytics

Mumbai

2025

Abstract

This research project examines how mathematical optimization techniques are enhancing the operational efficiency and customer satisfaction of contemporary tiffin (meal) delivery services. The study investigates key optimization concepts and their practical applications in improving route planning, resource allocation, real-time scheduling, and cost reduction in the tiffin delivery business. By combining advanced mathematical models with current delivery technologies, the research aims to demonstrate how optimization plays a crucial role in improving service quality and ensuring sustainable growth for meal delivery enterprises.

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1 Introduction and Literature Review

1.1 Industry Context

Tiffin delivery services constitute a vital component of urban food ecosystems, offering freshly cooked, home-style meals primarily to office-goers, students, and busy households. Especially prevalent in metropolitan cities like Mumbai, Delhi, and Bangalore, tiffin services face increasing operational challenges due to rising demand, urban traffic congestion, dynamic customer preferences, and cost pressures. While traditional models, such as the Mumbai Dabbawalas, have achieved six-sigma efficiency through manual, highly structured logistical networks [1, 4], the emerging unorganized sector of individual home chefs lacks such cohesive systems [3]. To address these challenges, mathematical optimization has emerged as a powerful tool for streamlining service delivery and managing resources effectively [6].

1.2 The Need for Optimization

Optimization techniques enable tiffin providers to allocate limited resources such as delivery personnel, kitchen equipment, and ingredients more efficiently. They also help design delivery routes that minimize travel distance and time, thereby reducing costs and improving punctuality [5]. Furthermore, heuristics and real-time scheduling algorithms support adaptive order management, allowing businesses to accommodate last-minute changes and ensure consistent service quality in a dynamic urban environment.

This study seeks to explore these optimization frameworks in the context of tiffin delivery. The main objective is to model and analyze how these mathematical strategies can overcome logistical and operational inefficiencies to enhance customer satisfaction and business sustainability.

1.3 Key Concepts of Optimization Investigated

1.3.1 Linear Programming (LP)

Linear programming forms the foundation for solving many resource allocation tasks in tiffin delivery. LP models aim to maximize or minimize a linear objective, such as minimizing the total delivery cost, while complying with resource constraints like delivery personnel availability and kitchen capacity. LP helps businesses make decisions on the number of batches to prepare, staff scheduling, and allocation of delivery routes to different drivers [8].

1.3.2 Integer Programming (IP)

In real-world tiffin services, many decisions are discrete, such as assigning specific delivery staff to particular routes or selecting delivery time windows. Integer programming, an extension of LP, handles these integer variables [11]. IP models are used to solve vehicle and personnel assignment problems, ensuring efficient distribution without operationally impractical fractional allocations.

1.3.3 Vehicle Routing Problem (VRP)

The Vehicle Routing Problem lies at the core of optimizing multi-stop food delivery routes. VRP seeks the minimal total travel distance or time for multiple delivery vehicles while satisfying constraints like vehicle capacity and delivery time windows [2]. For tiffin services with multiple routes and drivers, VRP algorithms allow assigning optimized routes that reduce fuel consumption, labor costs, and delivery delays. Advanced versions incorporate traffic and dynamic customer inputs.

1.3.4 Scheduling Algorithms and Heuristics

Scheduling concerns optimizing meal preparation timelines in the kitchen and syncing it with delivery times. Algorithms in this category ensure that meals are prepared just in time for dispatch, reducing waiting and spoilage risks. Due to the complexity and scale of delivery problems, exact solutions may be computationally expensive. Heuristic strategies and meta-

heuristic approaches like genetic algorithms provide efficient, near-optimal solutions within practical time limits, balancing solution quality with the computational effort critical for daily dynamic operations.

1.4 Areas of Application

1.4.1 Optimizing Delivery Routes

In urban environments, efficient routing is crucial [5]. VRP techniques plan routes for each delivery vehicle or person, taking into account meal drop-off points, kitchen locations, and vehicle capacity. Dynamic routing can also modify plans daily based on real-time traffic updates and new orders.

1.4.2 Resource Allocation and Dynamic Scheduling

Allocation models optimize the assignment of cooking resources and delivery personnel. Linear and integer programming frameworks help balance workloads across kitchens, schedule personnel shifts according to peak demand, and forecast ingredient needs to minimize inventory costs and wastage.

1.4.3 Cost Minimization and Customer Satisfaction

By integrating route optimization with resource management, tiffin businesses can target cost reduction. Models identify the best routes that minimize fuel and labor costs, suggest optimal staff numbers needed per shift, and estimate ingredient purchases to decrease excess inventory. Combined, these efforts improve profitability without compromising service quality.

2 Methodology

This research undertakes a multi-step approach combining literature review, mathematical modeling, simulation, and comparative analysis to develop a holistic solution for the unorganized tiffin sector.

2.1 Research Approach

- **Literature Review:** A comprehensive review of academic papers, case studies, and industry reports on tiffin delivery logistics and optimization methods was conducted to build foundational knowledge.
- **Mathematical Formulation:** Key operational problems, specifically customized meal packaging and delivery routing, were formulated using Dynamic Programming and heuristic frameworks.
- **Prototype Development:** A functional full-stack prototype was developed to simulate the application of these models in a user-facing environment.

2.2 Data Acquisition and Database Schema

To validate the platform’s search, filtering, and algorithmic capabilities, a representative repository of tiffin service providers was curated. This structured dataset serves as the foundation for testing matching logic and ensures that early adopters have immediate access to diverse options. The data, presented in Table 1, includes critical operational parameters such as cuisine type, dietary classification, and pricing tiers.

Table 1: Representative Tiffin Service Provider Database

ID	Provider Name	Location	Cuisine	Type	Rating	Price
1	Gupta’s Kitchen	Andheri West	North Indian	Veg	4.5	3200
2	Sharma’s Meals	Andheri West	Home-Style	Veg	4.8	3500
3	Khan’s Mughlai	Bandra West	Mughlai	Non-Veg	4.6	4500
4	Patel’s Gujarati	Dadar West	Gujarati	Veg	4.7	3000

ID	Provider Name	Location	Cuisine	Type	Rating	Price
5	Fernandes Goan	Borivali West	Goan	Both	4.4	4200
6	South Spice	Matunga	South Indian	Veg	4.9	2800
91	The Dabba Co.	Versova	Multi-Cuisine	Both	4.5	3700
92	Taste of Home	Malad West	East Indian	Non-Veg	4.8	4800
93	Tiffin Wallah	Grant Road	Mughlai	Non-Veg	4.7	4500
94	Rajasthani Rasoi	Thane West	Rajasthani	Veg	4.6	3400
95	Konkan Coastal	Dadar West	Malvani	Both	4.8	5100

Note: The table above represents a meaningful extract of the larger dataset used for simulation purposes.

3 Mathematical Optimization Algorithms

To address the operational inefficiencies identified, we developed two core algorithms: a meal package customization algorithm using Dynamic Programming, and a route recommendation system using a greedy heuristic.

3.1 Algorithm 1: Meal Package Customization

This algorithm addresses the need for customers to build customized meal plans within a fixed budget while maximizing subjective value (e.g., nutritional content or user rating). It is formally defined as a variant of the 0/1 Knapsack Problem with an equality constraint on the number of items selected [7]. This approach is related to broader assortment optimization problems [10] and customized allocation models [11].

3.1.1 Problem Formulation

Let N be the total number of available meals after initial filtering. For each meal i (where $i = 1, 2, \dots, N$), we possess:

- p_i : The price of meal i .
- v_i : The value associated with meal i .

The inputs include a maximum budget B and a required number of meals M . We define a binary decision variable x_i :

$$x_i = \begin{cases} 1 & \text{if meal } i \text{ is selected} \\ 0 & \text{otherwise} \end{cases}$$

The objective is to maximize the total value Z :

$$\text{Maximize } Z = \sum_{i=1}^N v_i x_i$$

Subject to the following constraints:

$$\sum_{i=1}^N p_i x_i \leq B \quad (\text{Budget Constraint}) \quad (1)$$

$$\sum_{i=1}^N x_i = M \quad (\text{Cardinality Constraint}) \quad (2)$$

$$x_i \in \{0, 1\} \quad \forall i = 1, \dots, N \quad (3)$$

3.1.2 Dynamic Programming Solution

We employ a Dynamic Programming (DP) approach to solve this NP-complete problem efficiently for typical input sizes. The state $dp[m][b]$ is defined as the maximum value obtainable by choosing exactly m meals with a total cumulative price of at most b .

The recurrence relation used to populate the DP table is derived from the standard Knapsack problem [7]:

$$dp[m][b] = \max(dp[m][b], \quad v_i + dp[m-1][b-p_i])$$

This relation evaluates the optimal sub-structure by either excluding meal i (inheriting the previous best value for state $[m][b]$) or including meal i (adding its value to the optimal solution for $m-1$ meals with the remaining budget $b-p_i$).

3.2 Algorithm 2: Greedy Route Recommendation

For delivery logistics, specifically for independent providers lacking centralized dispatch systems, we propose a Greedy Route Recommendation heuristic. This approach prioritizes immediate utility at each decision node, suitable for real-time, decentralized decision-making.

3.2.1 Mathematical Definitions

The delivery environment is modeled as a graph $G = (V, E)$, where V represents customer locations and E represents the paths between them. Let:

- D_{max} be the total available distance budget for a delivery run.
- $d(u, v)$ be the traversal cost (distance or time) between nodes u and v .
- $U(v)$ be the utility score of visiting node v (e.g., based on order urgency or delivery window proximity).

3.2.2 Selection Heuristic

At any current node u , with a cumulative distance traveled D_{total} , we first identify the set of feasible candidate nodes $C(u)$:

$$C(u) = \{v \in V \mid (u, v) \in E, v \notin R_{visited}, D_{total} + d(u, v) \leq D_{max}\}$$

This set $C(u)$ filters for nodes that are directly reachable, have not yet been visited during the current run, and whose traversal will not violate the total distance budget D_{max} .

The next node v_{next} to visit is selected by maximizing immediate utility from the candidate set:

$$v_{next} = \operatorname{argmax}_{v \in C(u)} U(v)$$

This greedy selection process repeats until $C(u)$ is empty. While not guaranteed to find the globally optimal route (which would require solving the NP-hard VRP), this heuristic provides a computationally lightweight solution for intra-day operational adjustments.

3.2.3 Algorithm: Greedy Route Recommendation

Algorithm 1 Greedy Route Recommendation

```
1: Route  $\leftarrow$  [ ]
2: CurrentNode  $\leftarrow$  StartNode
3: TotalDistance  $\leftarrow$  0
4: while TotalDistance < MaxDistance do
5:   candidates  $\leftarrow$  [ ]
6:   for NextNode in Graph.neighbors(CurrentNode) do
7:     if NextNode  $\notin$  Route and TotalDistance + dist(CurrentNode, NextNode)  $\leq$ 
       MaxDistance then
8:       candidates.append(NextNode)
9:     end if
10:  end for
11:  if candidates is empty then
12:    break
13:  end if
14:  BestNode  $\leftarrow$  pickbest(candidates)
15:  Route.append(BestNode)
16:  TotalDistance  $\leftarrow$  TotalDistance + dist(CurrentNode, BestNode)
17:  CurrentNode  $\leftarrow$  BestNode
18: end while
19: return Route
```

4 Results and Discussion: Prototype Implementation

The development of the NextMeal functional prototype served to validate the theoretical models in a simulated operational environment. The focus was on abstracting complex logistical data into an accessible user interface (UI) to facilitate adoption by non-technical stakeholders.

4.1 User Acquisition and Onboarding Interface

The initial user engagement layer (Landing Page) was designed to establish immediate value proposition clarity.

- **Visual Hierarchy:** The hero section utilizes high-fidelity imagery coupled with a concise value statement ("Homely Meals, Delivered") to reduce bounce rates among the target demographic.
- **Process Abstraction:** The complex backend operations are distilled into a three-step linear user flow (Choose Plan → Find Provider → Enjoy Delivery) presented on the landing page, reducing cognitive load for new users.
- **Product Transparency:** To address trust deficits in the unorganized sector, the interface explicitly visualizes standardized meal compositions pre-purchase.

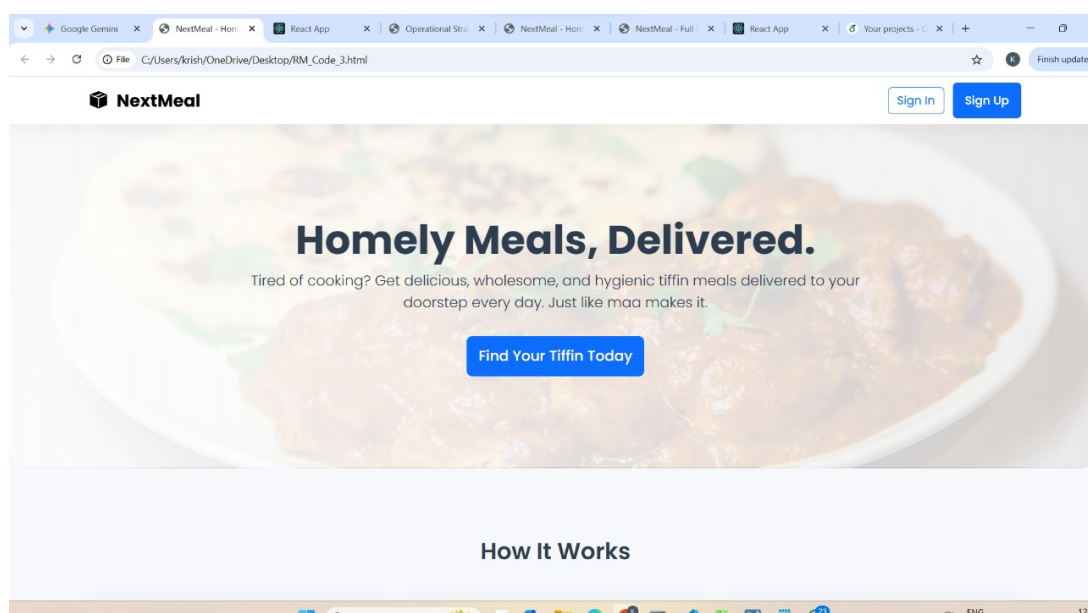


Figure 1: The NextMeal landing page, establishing the value proposition.

4.2 Role-Based Access Control (RBAC)

The system utilizes a unified authentication portal with role-based branching. A distinct toggle mechanism during the login phase directs users to either the Customer Flow (Discovery Dashboard) or the Provider Flow (Management Dashboard). This unified architecture reduces development overhead while maintaining strict data segregation between stakeholder types.

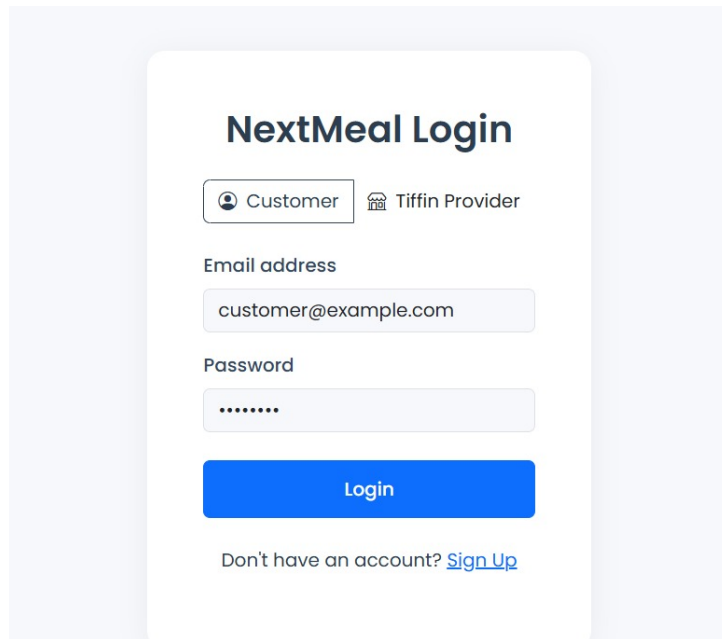
The image shows a login form titled "NextMeal Login". At the top, there are two toggle buttons: "Customer" (selected) and "Tiffin Provider". Below these are input fields for "Email address" (containing "customer@example.com") and "Password" (masked with dots). A blue "Login" button is positioned below the password field. At the bottom, there is a link that says "Don't have an account? [Sign Up](#)".

Figure 2: Unified login portal with role-based access for Customers and Tiffin Providers.

4.3 Constraint-Based Discovery Dashboard

Post-authentication, customers interact with the primary Discovery Dashboard. This interface is driven by a constraint-filtering engine designed to narrow the search space effectively.

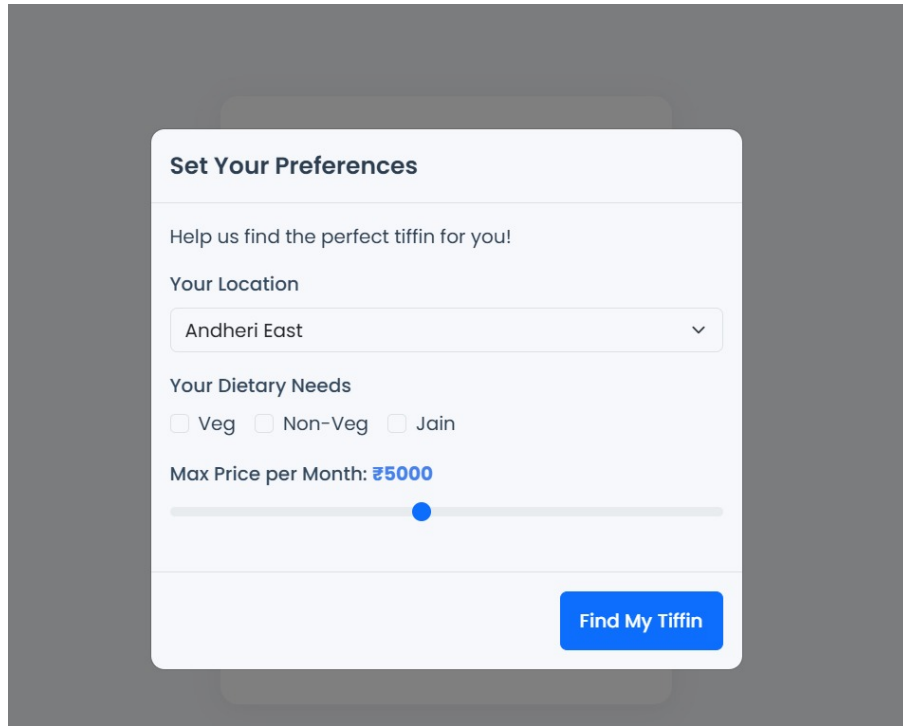
4.3.1 Preference Capture Modal

Upon initial access, a modal interface captures critical user constraints:

- **Geospatial Constraints:** Location data (e.g., "Andheri East") is captured to ensure hyperlocal relevance in search results.
- **Dietary Constraints:** Boolean filters for specific dietary requirements (e.g., Vegetarian, Jain) act as hard constraints in the database query, instantly excluding incompatible

providers to prevent user frustration.

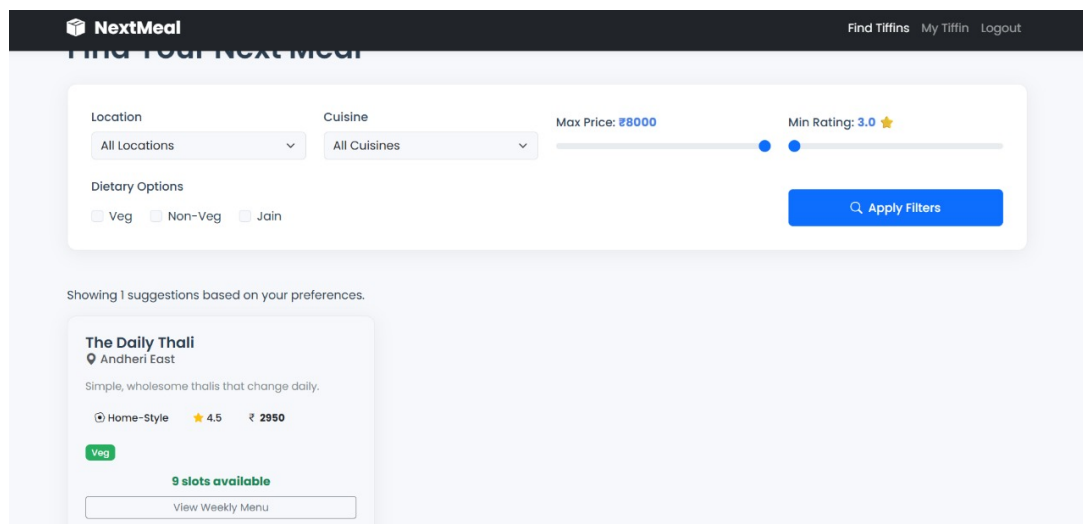
- **Economic Constraints:** A sliding scale input allows users to define a maximum monthly budget, filtering out providers that exceed their financial thresholds.



The image shows a modal titled "Set Your Preferences" with a light blue background. It contains the following elements:

- A heading "Set Your Preferences" followed by the text "Help us find the perfect tiffin for you!"
- A section "Your Location" with a dropdown menu showing "Andheri East" and a downward arrow.
- A section "Your Dietary Needs" with three checkboxes: "Veg", "Non-Veg", and "Jain".
- A section "Max Price per Month: ₹5000" with a horizontal slider bar and a blue dot indicating the selected price.
- A blue button at the bottom right labeled "Find My Tiffin".

Figure 3: Initial preference capture modal for filtering by location, diet, and budget.



The image shows a dashboard for "NextMeal" with a dark header. The main content area has a light blue background and contains the following elements:

- A header bar with the "NextMeal" logo and navigation links: "Find Tiffins", "My Tiffin", and "Logout".
- A filter section with the following controls:
 - "Location": A dropdown menu showing "All Locations".
 - "Cuisine": A dropdown menu showing "All Cuisines".
 - "Max Price: ₹8000": A horizontal slider bar with a blue dot.
 - "Min Rating: 3.0 ★": A horizontal slider bar with a blue dot.
 - "Dietary Options": Three checkboxes: "Veg", "Non-Veg", and "Jain".
 - A blue button labeled "Apply Filters" with a magnifying glass icon.
- A section titled "Showing 1 suggestions based on your preferences." containing a card for "The Daily Thali".

The "The Daily Thali" card includes the following information:

- Location: "Andheri East"
- Description: "Simple, wholesome thalis that change daily."
- Category: "Home-Style" with a 4.5 star rating and a price of ₹2950.
- Dietary Option: A green badge labeled "Veg".
- Availability: "9 slots available".
- Button: "View Weekly Menu".

Figure 4: The constraint-based discovery dashboard displaying filtered search results.

4.3.2 Customer Subscription Dashboard

Upon selection and subscription, the customer is directed to a personal dashboard. This interface provides real-time status updates on daily meals (e.g., "Out for delivery," "Preparing"), a view of the full weekly menu, and access to subscription management tools, such as pausing the service or contacting the provider. This transparency is critical for building long-term trust and customer retention.

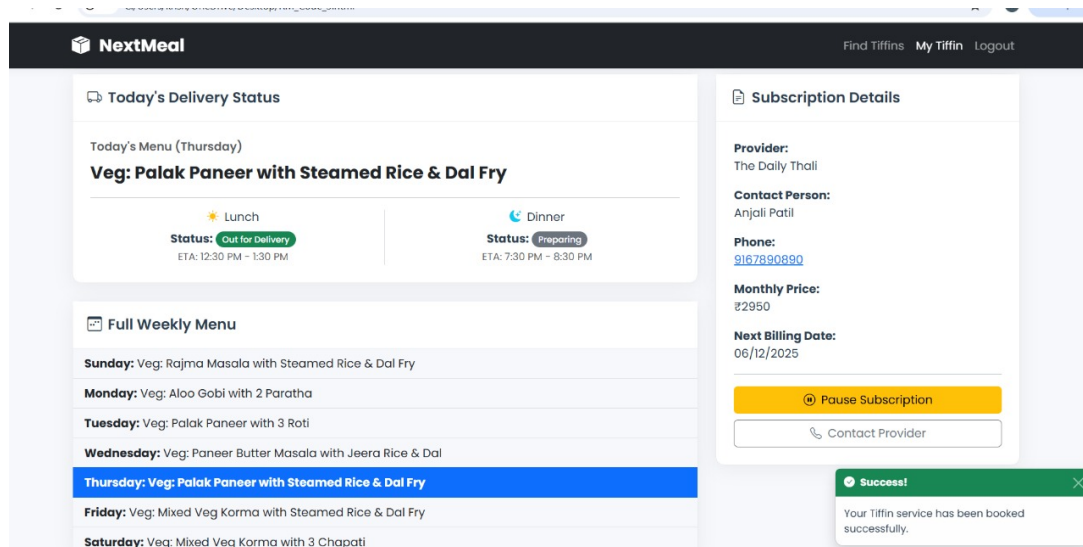


Figure 5: The customer-facing dashboard showing delivery status, weekly menu, and subscription details.

4.4 Supply-Side Operational Metrics

For service providers, the interface shifts focus from discovery to operational management. The Provider Dashboard aggregates disparate order data into high-level Key Performance Indicators (KPIs).

- **Active Subscriber Count:** Provides immediate visibility into current demand levels.
- **Revenue Projections:** Displays estimated monthly earnings based on active subscriptions, aiding in financial planning for micro-entrepreneurs.

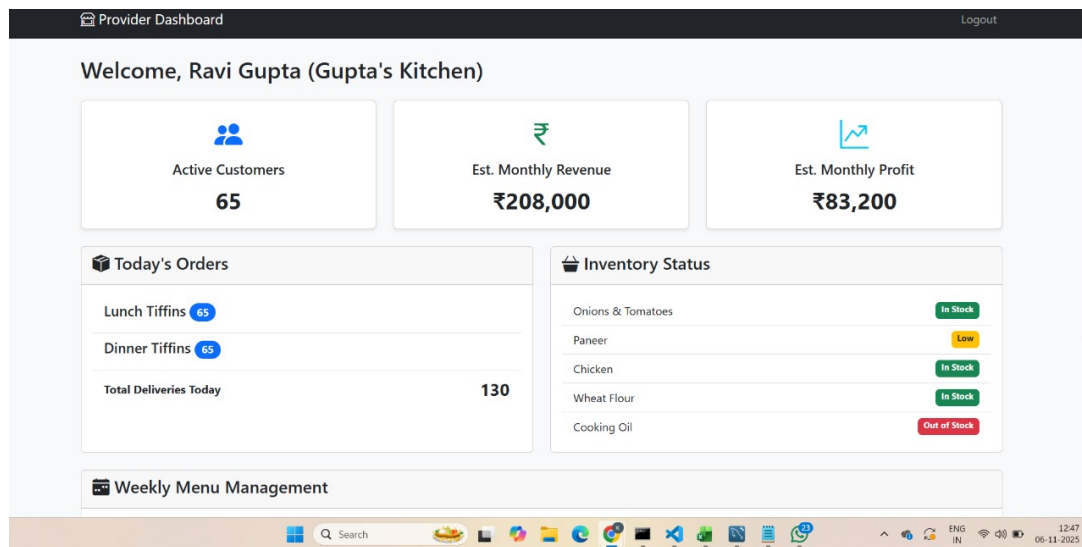


Figure 6: Provider-side management dashboard aggregating key performance indicators (KPIs) such as active customers, revenue, and inventory status.

5 Conclusion and Future Scope

5.1 Conclusion

This project successfully demonstrated the viability of digitizing the fragmented tiffin service sector through mathematical optimization and modern web technologies. The prototype confirms that complex logistical parameters can be effectively abstracted into a user-friendly interface. By integrating a structured relational database with heuristic optimization algorithms, we have established a foundational framework for a scalable ecosystem that balances customer convenience with the operational sustainability of small-scale providers.

5.2 Future Scope: Transition to Production

While the current prototype validates the core concepts, full-scale commercial implementation requires significant infrastructure enhancements.

5.2.1 Database Persistence and Scalability

The current transient state management must be replaced by a persistent, ACID-compliant Relational Database Management System (RDBMS) such as PostgreSQL. This is critical for maintaining long-term user histories, secure transaction logs, and managing the complex many-to-many relationships between customers, varied menus, and providers.

5.2.2 Full-Stack Architecture Upgrade

To support high concurrency and robust security, the application logic will be migrated to an enterprise-grade backend framework, such as Django (Python). This will handle authentication, data validation, and complex business logic. Concurrently, the frontend will be rebuilt using a reactive framework like React.js to ensure low-latency user interactions and efficient state management.

5.2.3 Integration of Artificial Intelligence

Future iterations will incorporate machine learning models to enhance platform intelligence:

- **Personalization Engine:** Collaborative filtering algorithms will analyze historical user rating patterns to automatically curate personalized menu recommendations, moving towards frameworks similar to those described in [9].
- **Demand Forecasting:** Time-series analysis of order history will be utilized to predict future ingredient requirements, enabling providers to optimize procurement and minimize perishable inventory wastage.

5.2.4 Centralized Logistical Optimization

The long-term roadmap includes moving beyond individual route recommendations to a centralized dispatch model. By aggregating deliveries from multiple providers within shared geographical clusters, advanced Vehicle Routing Problem (VRP) algorithms can be deployed to optimize shared delivery fleets, significantly reducing aggregate fuel consumption and carbon footprints.

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